# AST 2000 - Part 2
# Planetary Orbits

Welcome to Part 2 of the AST2000 Spacecraft Project. There are 3 components in this part, one of which is optional for those working alone: To begin, you will need to simulate the orbits of your planets and your star. Next (this is optional for those working alone), you will test your orbits using Kepler's laws. Finally, you will take a look at data of the motion of another group's star and see if their planets are observable from looking at their Sun's wobbling. **For this part you are only supposed to deliver a partial paper including: (1) abstract, (2) a results section, (3) a discussion and conclusions section (4) figures for these. No introduction or method section is expected for this part. Remember to also deliver a hand-written flow chart for your code.**

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## GOALS

- ∇ Plot the orbits of the planets in your solar system.
- ∇ Simulate the orbits of the planets in your solar system.
- ∇ Use Kepler's laws to verify the integrity of your planetary orbits (optional for those working alone).
- ∇ Analyse how your planets affect their star and evaluate whether the effect is noticeable from other solar systems.

## SIMULATING THE ORBITS

To reduce our computational load we introduce the following simplifications:

- ⊗ Ignore planet-planet interactions and only consider the gravitational pull on the planets by the star.
- ⊗ Ignore relativistic effects.
- ⊗ Ignore the gravitational pull on the star by the planets. Assume the star is fixed at the origin throughout the simulation.
- ⊗ We are assuming that all the orbits exist in the same $x$-$y$ plane. You can therefore safely ignore the $z$-axis.
- ⊗ All planets both orbit the star *and* rotate about the $z$-axis in a counterclockwise motion.
- ⊗ Ignore all other astronomical bodies in the solar system, including moons.

## CHALLENGES

Relevant theory for Part 2 is described in Lecture Notes 1B and Lecture Notes 1C which you can find on the semester page. Make sure you have read them *before* you continue.

In addition to the lecture notes you should also check the integration section in the Numerical Compendium. Note that the Leap Frog integration scheme is energy conserving.

### A. Planetary orbits

We encourage you to use AU units to prevent floating point overflow. Also, the gravitational constant is very simple in AU units: [**?** ]

$$G = 4\pi^2 \text{ AU}^3\text{yr}^{-2}\text{M}_\odot^{-1}$$

#### *1. Plotting the Orbits*

Your first objective is to use the analytic solution of elliptical orbits found in Lecture Notes 1B in order to plot the orbits of all your planets. You can access all necessary data in the `SolarSystem` object held by your instance of the `SpaceMission` class.

#### *2. Simulating the Orbits*

Now that you have plotted the planetary orbits of your solar system, the next step is to simulate the orbits. "Wait, didn't we just find the analytic solution?" I hear you say. Well, while the analytic solution provides us with the shape of the orbits, it *does not* describe the position of the planets as a function of time $t$. We need to find this information by simulating their motion.

Simulate the planetary orbits for at least 20 revolutions of your home planet. Use a minimum of 10,000 time steps per year to ensure your results are accurate. Depending on your solar system you may even need more. Comment on your chosen number of time steps: why is this number sufficient? To verify your simulation has been completed successfully, plot your simulated orbits on top of the analytic orbits (the accuracy of your orbits will be checked later).

Running the simulation may take a considerable amount of time. **Remember to save your orbits**

so you only have to perform this simulation once. You should also review your options and choose a programming style that is fast and that you understand well. If you have no preference, we invite you to take a look at the Numerical Compendium.

### B. Comparing your Orbits to Kepler's Laws

*This is optional for those working alone.*

1. Determine the area swept out by your planet in some time interval $\Delta t$.
   (a) Compare two areas: one where the planet is close to the aphelion and one where the planet is close to the perihelion. To what degree are the areas equal?
   (b) What distance did the planet travel as it swept out these two areas?
   (c) What mean velocity did the planet travel with?
2. Check if all your orbits are consistent with Kepler's 3rd law. Is there a measurable difference between Kepler's original version and Newton's corrected version?

### C. Can Extraterrestrials Discover the Planets in your Solar System?

We are now going to take a second look at our planetary orbits. This time however, you are going to include the gravitational pull on your star by its planets.

Be careful not to confuse your orbits later in the project; **the orbits you simulate for this challenge should not be used in anywhere else.**

Furthermore, half of this challenge involves the exchange of data between you and another group. Make sure you have someone to cooperate with!

#### *1. The Solar Orbit*

1. Choose one of the planets in your solar system: preferably one of the largest and as close to the star as possible (to maximize gravitational pull). Place the star at the origin for $t = 0$ and let its initial velocity be zero. Doing this places you in a frame of reference where the center of mass drifts; find a way to change your frame of reference to the center of mass frame.

   Ignore all other planets and simulate the orbits of the star and your chosen planet using your favorite integration method. Choose your integration parameters based on your experience with simulating the orbits.

2. With the solar orbit in hand we are now going to test its accuracy using an analytic approach. Your first task is to study the two-body system in general, then use your findings in order to study your specific solar orbit.
   (a) Show that the total energy of the two-body system as seen from the center of mass frame can be written as

   $$E = \frac{1}{2}\hat{\mu}v^2 - \frac{GM\hat{\mu}}{r} \tag{1}$$

   where $v = |\dot{\mathbf{r}}|$ is the relative speed, $r = |\mathbf{r}|$ is the relative distance, $\hat{\mu}$ is the reduced mass and $M$ is the total mass of the system.
   (b) Use equation (1) to test the numerical accuracy of your new orbits. Is the energy conserved to a reasonable degree?

#### *2. The Radial Velocity Curve*

1. The radial velocity curve of a star is dependent on three parameters: the peculiar velocity of the center of mass, line of sight, and the inclination of the solar system from the line of sight.

   Choose an arbitrary set of parameters and design the radial velocity curve for your star based on your newly simulated solar orbit. With the exception of inclination angles $i \approx 0$, you are free to choose any physically consistent values for your parameters.

   Now add some independent Gaussian noise to the data. Let the mean noise be zero and the standard deviation be 1/5 of the maximum value of your velocity curve.

2. Contact the other group you are cooperating with and exchange your radial velocity curves (as well as your star masses which you need). Your task is to analyse their velocity curve and provide an estimated lower boundary for the mass of the planet in their solar system. You should only use the graph and read of values by-eye. In your report, show clearly how you read of the plot and why in this manner. However, before you can find the mass you need to adjust the data for the star's peculiar velocity by removing any existing drift.

3. Once both groups have finished their analyses, compare your results to the actual masses. Do you expect to find exact numbers? Why, why not?

#### *3. The light curve*

*This is optional for those working alone.*

1. Let the inclination of your solar system as seen from an outside observer be $i = 90°$. You need to design an approximate light curve, remember that the received light depends on the area of the planet in front of the star. That is, you do not need to estimate the actual flux from your star; only the relative change in the flux, caused by the planets eclipsing the star. Define your flux relative to the maximum flux emitted by your star (such that flux = 1 when no planet is eclipsing the star).**To make this task easy, estimate the relative flux with the full planet's shadow, then make a straight line connecting the relative flux without planet (equal to 1) with the part of the light curve when the planet is fully in front of the star.**

   Again, add some independent Gaussian noise to the data. Let the mean noise be zero and the standard deviation be $10^{-4}$ relative flux. Do you think it is possible to detect your planet?

2. Contact the other group you are cooperating with and exchange light curves. Your task is to analyse their light curve and attempt to estimate the radius and density of the planet eclipsing their star. In order to do this, you need data from your analysis of the same other group's radial velocity curve.

3. Once both projects have finished the analysis, compare your results to the actual radii and densities.

#### *4. The radial velocity curve with more planets*

*This is optional for everybody. If you do this task, we will add the points to the total score before rounding up.*

1. You are now going to revisit your simulation of the solar orbit and include 2-4 additional large planets from your solar system. The simulation is in principle exactly the same as before, but you may have to redesign your integration loop depending on how you structured your initial program.

2. Again, choose an arbitrary (yet physically plausible) set of parameters and design a new radial velocity curve. Exchange this curve with the other group you are cooperating with and attempt to resolve the number of planets the other project included in their simulations. Once both groups have made their guesses, compare with the actual number of planets in the simulations.

## VERIFICATION/CONFIRMATION

The Part 2 challenges are now complete! All that remains now is to verify the numerical accuracy of your orbits. Check your orbits using the `verify_planet_positions` method from the `SpaceMission` class. Your calculations are expected to be within a 1% margin.

Note that the orbits you need to verify are the original orbits where the star is *stationary* (i.e. the orbits from challenge A).

## VISUALISING THE SOLAR SYSTEM

Use the `generate_orbit_video` method from the `SpaceMission` class to generate an XML file containing your numerically simulated orbits. Assuming you have downloaded MCAst, place the XML file in the `data` directory within the MCAst installation directory. You can now explore your solar system using the SolarSystemViewer software (Note that you need to start the application called `SSView` in the MCAst directory for this, not `MCAst`). Does it live up to your expectations?



FIG. 1. Orbital motivational duck.

# AST 2000 - Part 2
# Tips, Hints & Guiding Questions

## I. IN GENERAL

→ Note that the first challenge is the most important challenge in Part 2. Without the planetary orbits you are effectively incapable of continuing the project.

→ Be extra careful with your units! We actually mean it when we *"encourage you to use AU units"*.

## II. HINTS FOR SOME OF THE CHALLENGES

### A. The Gravitational System

Since the orbits are decoupled (they do not interact), each orbit is described by the same differential equation:

$$m_p \ddot{\mathbf{r}} = -G\frac{m_s m_p}{|\mathbf{r}|^3}\mathbf{r}, \tag{a}$$

Here, $m_s$ is the mass of the star, $m_p$ is the mass of the planet and $\mathbf{r}$ is the position vector of the planet. Initial conditions for the planetary orbits are accessed through the `SolarSystem` object held by your `SpaceMission` instance.

You are most likely better off simulating all your planets' trajectories simultaneously using NumPy arrays. The numerical accuracy of your calculations are really important here; we advise the use of higher order integration methods such as the Leap Frog scheme. You can find more hints in the Numerical Compendium.

### B. Simulating the New Orbits

How does the the gravitational system differ from the original simulation? Is equation (a) still valid for any or all bodies? What has changed?

To change your frame of reference to the center of mass frame, you need to construct the position vectors of the bodies as seen from the center of mass. How do the orbits look from the center of mass? Can you calculate the position vector of the center of mass as seen from your simulation's frame of reference?

### C. Designing the Radial Velocity Curve

Here you need to use your solar orbit from challenge C 1 as seen from the center of mass frame. Your first task is to add an arbitrary peculiar velocity to the center of mass. Next comes the inclination of the orbit; you should be able to deduce the effect of tilting the orbital plane using simple trigonometry. Last but not least, you need to choose your line of sight. To guide your intuition, take a look at figure 2: You can think of the line of sight as placing the stick-man somewhere on the $(x, y)$ plane, recalling that he is only able to observe the radial velocity along his line of sight. The easiest cases to implement are when the line of sight is either along the $x$-axis or the $y$-axis. Can you explain why this is the case?

Unless you have a strong passion for a specific programming style, I would like to personally recommend checking out `JIT`! It's fast and simple. Otherwise standard `NumPy` vectorization is key here. Regardless of your chosen style, be careful to mix `JIT` and `NumPy`, check out the $\zeta$ Half-Life example in the Numerical Compedium.

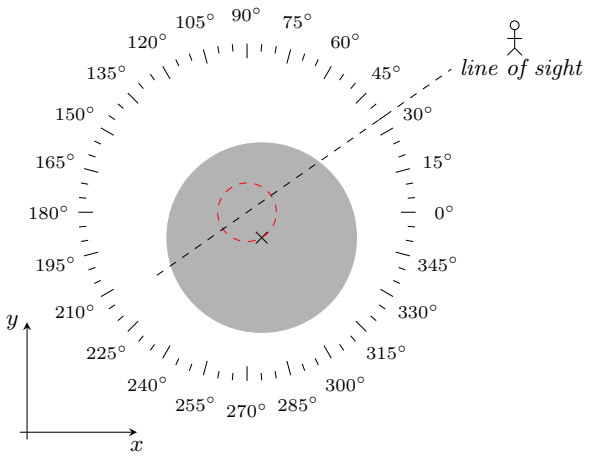



FIG. 2. An illustration of an observer's line of sight to a distant solar system. The star (gray disk) follows its circular orbit (dashed red line) in the $(x, y)$ plane. An observer (stick-man) sees the star system from an arbitrary position. Note that in this illustration the inclination angle $i = 90°$, this is for illustrative purposes only.